Passive Port Certificates and Their Spectral Boundary

A Source-Specific Ram Calculus Report on Manuscript M524

Dr. Muralidhara Subbarao
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Abstract

We audit the pinned manuscript *Pure-Point Spectrum for the Two-Dimensional Anderson Model at Every Positive Disorder*, separating its finite passive matrix identities from its weighted probability, two-dimensional capacity, fractional-moment, and spectral-limit arguments. We prove a heterogeneous passive port-closure formula with an exact positive dissipation decomposition, its compositional and inexact-solve certificates, a physical-metric reduction and residual-to-observed-response bound, an exact barycentric energy decrement for conditional ideal Cauchy closure, and a centered-variance refinement for its actual cutoff law. These finite results apply to the manuscript's projected impedance, duplicated site ports, and probe closure without requiring a bound on the entire exterior inverse when the residual has the specified absorbed-source representation. We also prove a conditional finite-to-infinite resolvent-column lemma and identify why neither a finite inverse nor analyticity establishes its probabilistic hypothesis. The private Ram passages supply representation and error-accounting contracts; Schur, Woodbury, Ward, Galerkin, and rank-one spectral techniques are classical, and no historical novelty or independent verification of the headline localization theorem is claimed. Contribution assessment: C3/5, exact local loss certificates and a centered conditional closure refinement strengthen the pinned finite-port comparisons while the all-disorder multiscale and infinite-volume localization dependencies remain unverified.

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1 Pinned Problem and Claim Boundary

1.1 Identity, spaces, and quantifiers

The catalogue row is integer ID 524, with applicability L5, at the requested catalogue pin

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The source is dated September 23, 2026; this report's study date is October 7, 2026. The local source directory is the one in Appendix A. The checkout is a file snapshot without a `.git` repository: the catalogue's commit association is recorded, not represented as a locally verified Git HEAD. All fourteen local source files match the stored SHA-256 manifest and Git blob SHA-1 tree records; this is local manifest consistency, not an independent remote commit-to-tree verification. Content hashes identify the actual files read. Applicability L5 is not the contribution rating.

For each *fixed* $h > 0$, the probability space is $\Omega_h = [-h, h]^{\mathbb{Z}^2}$ with its product Borel sigma algebra and product of normalized Lebesgue measures $(2h)^{-1}\,\mathrm{d}v_x$. On the complex Hilbert space $\ell^2(\mathbb{Z}^2)$ with counting measure and inner product conjugate linear in the first entry, define

$$(H_v u)(x) = \sum_{|x-y|_1=1} u(y) + v_x u(x), \qquad D(H_v) = \ell^2(\mathbb{Z}^2), \quad \|H_v\| \le 4 + h. \tag{1.1}$$

There is no continuum boundary condition. The finite global domains are $\mathbb{T}_N^2 = (\mathbb{Z}/3^N\mathbb{Z})^2$ with periodic nearest-neighbor adjacency. Local Dirichlet restrictions delete bonds exiting the chosen physical site set; they are not periodic restrictions. Distances are shortest coordinate max-norm

distances unless marked otherwise. The small degenerate tori are irrelevant to assertions for all sufficiently large N; no numerical experiment below uses ambiguous small-torus adjacency.

The source's `main:theorem` asserts that for each fixed h there is an event of probability one on which H_v has a complete orthonormal eigenbasis and spectrum $[-4-h, 4+h]$, with dense eigenvalues. It does *not* assert that every point of that interval is an eigenvalue, one common event for all h, uniform-in-energy decay constants, exponential decay of every eigenfunction, or dynamical localization. We record that assertion as the pinned problem, not as a theorem independently proved by this report.

1.2 Local matrix conventions

Physical sites and ports are separate finite sets. Several ports can share a site. The real matrix Y has columns $\sqrt{\sigma_j}e_{x_j}$, $\sigma_j > 0$. Thus $YY^\top$ adds the strengths at a site, but the port columns remain distinct. Zero-strength padded channels are permitted and give isolated identity entries in scattering matrices. For a real symmetric reference Hamiltonian $H^\#$ and real energy E, write

$$D = H^\# - E - iYY^\top, \qquad G = D^{-1}, \qquad S = I + 2iY^\top GY. \tag{1.2}$$

Reference values differ from true values only at open cell-port sites. All finite inverses are used on their nonsingular domains; almost-sure existence under the continuous diagonal law is not deterministic invertibility for every sample. Unless stated otherwise, norms are Euclidean operator norms. Superscripts $\top$ and $*$ mean transpose and Hermitian adjoint, respectively; they cannot be interchanged. Passivity means $-\operatorname{Im} D \succeq 0$, with $\operatorname{Im} D = (D - D^*)/(2i)$.

For real vector data $f_j \in \mathbb{R}^q$ the source energy is

$$\mathcal{Q}_S(f) = \frac{1}{2}\sum_{a,j} |S_{ja}|^2 |f_j - f_a|^2. \tag{1.3}$$

For symmetric unitary S and scalar $F = \operatorname{diag}(f)$ it equals $\operatorname{tr}(F^2 - FSFS^*)$. This trace representation uses unitarity; the pairwise definition remains meaningful for a merely contractive matrix, but the trace representation then needs a loss term.

2 Stepwise Source Dependency Audit

All substantive text and proofs in the eleven section files were read, as were the main file, bibliography, and README. There are no separately included mathematical appendices in this local manuscript. "Read" here does not mean that every estimate or quantifier has been independently proved or mechanically checked. The following audit specifies the interfaces actually encountered, including the narrow places where our results apply.

2.1 Finite algebra and resonant design

Matrix section. `mat:ward` proves $\operatorname{Im} G = G^*WG = GWG^*$, with $W = -\operatorname{Im} D \succeq YY^\top$, and passivity of Schur complements. For $W = YY^\top$, S is symmetric unitary. The proof of generic invertibility restores true diagonals on a kernel vector vanishing at ports, then uses the nonzero full-product monomial of a determinant. `mat:closure` obtains $S_z = A + Tz(I - zB)^{-1}T^\top$ through Woodbury. Its disk analyticity follows from $\|B\| \le 1$. `mat:energy-closure` requires that the open system and data do not expose the hidden closing value. The cutoff-Cauchy estimate `mat:cutoff` depends on cancellation of the conditional mean, not on a bound for the absolute value of the

closing multiplier. `mat:weak` requires dimension-independent weak-L^1 bounds, proved there through positive matrix measures and a Hilbert-valued kernel decomposition. We do not replace it with an operator-norm bound $(1-|z|)^{-1}$, which is useless on the circle. The phase lemma `mat:phase` and the piecewise-polynomial probe profile require $|\vartheta'|^2 \le C\vartheta$; endpoint vanishing is quadratic.

Resonant cells. `cell:geometry` uses a square with four deleted corner sites, root 0, punctured interior J, boundary B, and adjacency $M : \mathbb{R}^B \to \mathbb{R}^J$. On $P = (\ker M)^\perp$ its singular values lie in $[1, \sqrt{3}]$; each corner has the block $\left(\begin{smallmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{smallmatrix}\right)$ and one missing direction. The inverse-calibrated root is $c(v) = E + p^\top A^{-1} p$ for $A = \mathrm{Adj}_J + \mathrm{diag}(v - E)$. `cell:profile` obtains a polynomial inverse bound from a signed profile deformation whose weighted form derivative is positive. This is a separate design argument, not positivity of the original Hamiltonian. `cell:branch` introduces shell jitters, a common shift, and a cutoff-Cauchy hidden root with $\sigma = e^{-2\nu L}$. The branch likelihood is bounded by $\exp(CL^3 + CL^2 \log L)$ relative to the specified true law with the same corner boxes. The order of the exponents D_1, D_2, D_3 matters: the shell depth cannot be chosen using an error exponent which itself depends on that depth.

Weighted impedance. `imp:weighted` concerns arbitrary fixed complex symmetric passive exterior T and arbitrary source k, with

$$\rho = \left\| P(\mathcal{T} - iP)^{-1} k \right\|^2, \quad \mathcal{T} = T + \mathrm{diag}(v_i - i\tau a_i(v_i)).$$

It claims a polynomial inverse bound on a set carrying a fixed fraction of the ρ-*weighted* measure, uniformly in exterior norm and dimension. An unweighted high-probability bound is insufficient. The eight corner tests, polynomial norm comparisons, appreciable- and small-coupling cases, linewidth floor, and common-side phase integration are genuine prerequisites. At `imp:schur-interface`, eliminating the four missing directions transforms both $\widehat{T}$ and $\widehat{k}$. Section 4 supplies an exact loss-factor and residual certificate for this interface, not its probabilistic weight estimate.

Transmission. `trans:cell` claims

$$\mathbb{E}|t|^2 \ge e^{-C(\log L)^2} \mathbb{E}_B \rho, \qquad \mathbb{E}[|\mu||t|^2] \le e^{-cL} \mathbb{E}|t|^2.$$

Its proof separates the small-response region, a regular zero-jitter Schur inverse, a logarithmic-length resolvent-entry path, independent jitter propagation, and restoration of exponentially weaker artificial absorption. The normalized exterior source $k_* = (\widehat{T} - iI)^{-1}\widehat{k}$ avoids an assumed bound on $\widehat{k}$. The interior envelope `trans:envelope` sums over all internal port subsets to dominate adaptive choices. This is not a sparse-implementation complexity estimate: the displayed envelope can have $2^{|V|}$ terms.

2.2 Probability, scale limits, and the two-dimensional step

Hierarchy. `hier:sampling` samples true iid potentials first and attaches selector and likelihood-acceptance marks. It does not replace the physical law by the branch law. Coincident corner triples share the same box label. Conditioning must not reveal the numerical acceptance uniform at a hidden root; `hier:hidden` explains why that would truncate its conditional density. In `hier:cap`, a selected and accepted parent is an actual alternative in the current ensemble. The cap $Z_l(\theta) \le \theta K_l \mathbb{E}\mathcal{Q} + C\theta B_l$ removes an apparent $1/\theta$ endpoint divergence. Integrating $\min(2/\theta, K_l)$ costs $2 + 2\log(K_l/2)$, and the exponential cell gain must dominate this polynomial logarithmic cost. The admissibility condition on random data persists throughout both passes.

Screening. `screen:observation` is a probabilistic finite-dimensional kernel argument using all support patterns, not deterministic lattice unique continuation. `screen:projected` controls $\Pi^\top G \Pi$ for $\Pi = [J_g\ Y]$, rather than G. This distinction is precisely the observed-response distinction in Section 6. The full-stage and retained-wall path recursions are separate; the latter must remain uniform as wall spacing grows. `screen:limits` first fixes n, then chooses jump cutoff, local Dirichlet radius and hierarchy cutoff. Finite-range local approximants and offset-period averages justify spatial means. Only after taking the large-torus limit is $d_n \to d$ obtained from the one-sided almost-monotone estimate. No rate uniform in n is supplied.

Vanishing energy. `van:zero` is the decisive two-dimensional statement $d = 0$. The first variation uses a response-dependent sign whose measurability must respect all hidden values. A square-logarithmic voltage profile costs $C/\log M_0$; it bounds total probe transmission, while a separate wall argument bounds the unbounded affine weights on remote ports. The analytic function `van:analytic-function` uses an adjugate and determinant, not an analytically varying adjoint. The central lower bound needs near-unitarity in *trace norm* as well as a nonzero internal jump energy. The boundary moment requires the weak integral estimates from the matrix section. The support-preserving twist $v \mapsto v + s\vartheta(v)$ changes the product law and can alter ancestor validity gates; the source bounds both the density cost and gate exceptions. The order is: barrier constants, κ, M_0, cutoffs, C_0, a Cauchy tail cutoff and tolerance, then n, m, N. Finite passivity alone supplies none of this contradiction.

2.3 Bootstrap and spectral readout

Fractional moments. `boot:fractional` claims, for every fixed $0 < s < 1/16$, $h > 0$ and $E \in (-4-h, 4+h)$, constants $A, a > 0$ and N_0 such that for every $N \geq N_0$ and all torus sites

$$\mathbb{E}|(H_N - E)^{-1}_{xy}|^{s/2} \leq Ae^{-a\,\mathrm{dist}(x,y)}. \tag{2.1}$$

The proof uses mixture domination, a one-site L^2/L^1 bound under the transmission tilt, two buffered barriers, and independent inside/outside resampling. Independence is unconditional after dropping the original barrier indicator. The small return estimate uses $d_n \to 0$. Contraction takes the supremum over deterministic masks after the thresholds are fixed; no simultaneous good event for all masks is claimed. Removing two endpoint probes uses a 2×2 determinant identity and conditional fractional moments, with an exponent loss from s to $s/2$.

Spectral conclusion. `spec:column` uses (2.1), Fatou, a sample-dependent coordinate subsequence, and local equations to produce an ℓ^2 solution at each fixed real energy. It does not infer pointwise real-energy inverse convergence from ordinary nonreal strong resolvent convergence. `spec:averaging` is a rank-one spectral argument. `spec:readout` applies Fubini to the jointly measurable inverse-square spectral integral of the *true* operator, then averages one independent site value and intersects over countably many sites. A single site vector is not assumed cyclic for the full planar operator. Endpoint masses and the spectrum interval are addressed separately. Section 9 proves the conditional column bridge, with (2.1) explicitly left as an input.

3 Actual Private Ram Correspondences

The precise local paths and read ranges are recorded in Appendix A and the input ledger. No routing map is used as a substitute for the passages below.

Schur reconstruction and low-rank reuse. In *TR-core-pat-5*, subsection "Elimination is not pooling", `prop:schurresidual` and `eq:schurresidual` retain the interior forcing and show how an interior residual is transported to the retained rows. This maps literally to $\widehat{k}$ in `imp:schur-interface`, and to our (4.3). The theorem `thm:woodbury`, "Middle conditioning is necessary but not enough", and "Inexact actions and a robust middle test" require a checked absolute middle-matrix margin and the full updated residual. They motivate the feedback residual in Theorem 5.3. We do not attribute Woodbury inversion itself to Ram calculus.

Metrics and missing residuals. Volume 21, `ch01d-physical-metrics.tex`, equations `v21c01:physical-gram-definition`, `v21c01:residual-identity` and `v21c01:gram-equivalence`, distinguishes a synthesis metric, an omitted physical residual, and a relative Gram comparison. These passages motivate the nonorthogonal-coordinate result in Section 7. The source itself already uses real orthonormal boundary bases; adding a Gram matrix is needed only if one changes that representation. Calling a nonorthogonal coordinate matrix "symmetric" in its Euclidean norm would lose the physical meaning of passivity.

Analytic-domain discipline. *TR-rev-1-old-tr018*, "Analytic Continuation of the Ram Transform in Order ν", `thm:analytic-continuation` and `cor:interpolation`, requires common domains, fixed branches and pole-free inverse families, and warns that uniqueness of continuation does not give stability from noisy samples. Our closure parameter is z, not a fractional derivative order ν. The correspondence is the domain contract, not an identification of these different variables or a need to introduce fractional calculus into the Anderson model.

Finite-to-global limits. Volume 21, `ch01f-spectral-limits.tex`, contains a nested graph-core spectral-pollution example (`v21c01:pollution-proposition`), a strong-but-not-norm resolvent example (`v21c01:strong-not-norm`), and "Domain-aware approximate inversion" under `v21c01:stronger-contract`. These are explicit counterexamples to treating dense finite representations as spectral certificates. A finite-rank inverse cannot make the complementary infinite-dimensional identity a strict contraction. This explains the separate probabilistic and spectral contract below.

A correspondence deliberately not used. The core report's "Polynomial evaluation without eigendecomposition", `eq:filtererror`, assumes a self-adjoint operator on a declared real spectral interval. A complex symmetric passive D, or the compression B of a unitary scattering matrix, need not be normal. Its resolvent cannot receive that scalar Chebyshev error estimate without an additional theorem. Disk contraction bounds and physical residuals are used instead.

4 Schur Elimination With Exact Loss Accounting

Theorem 4.1 (Loss-factor Schur certificate)**.** *Let D be a finite complex matrix with $W = -\operatorname{Im} D \succeq 0$, partitioned into retained coordinates R and eliminated coordinates I. Assume D_{II} invertible. Set*

$$X = \begin{pmatrix} I_R \\ -D_{II}^{-1} D_{IR} \end{pmatrix}, \qquad D_e = D_{RR} - D_{RI} D_{II}^{-1} D_{IR}.$$

Then

$$DX = \begin{pmatrix} D_e \\ 0 \end{pmatrix}, \qquad D_e = X^* D X, \qquad -\operatorname{Im} D_e = X^* W X. \tag{4.1}$$

If $Y = \binom{Y_R}{0}$ *and* $W = YY^* + L^*L$, *then*

$$-\operatorname{Im} D_e = Y_R Y_R^* + (LX)^*(LX). \tag{4.2}$$

If D *is complex symmetric, so is* D_e. *For successive defined eliminations with lifts* X_1, X_2, *the total lift is* X_1X_2 *and the final loss factor is* LX_1X_2; *the same assertion holds for any finite number of defined stages.*

Proof. The second block row of DX is zero and the first is D_e. Premultiplication by X^* gives the middle identity in (4.1); taking Hermitian imaginary parts gives the last. Since $X^*Y = Y_R$, substitution of the factorization of W proves (4.2). Symmetry follows by transposing the Schur formula and using $(D_{II}^{-1})^\top = D_{II}^{-1}$. Applying the quadratic-form identity a second time gives $X_2^*X_1^*WX_1X_2$ and the asserted factor. Induction proves the finite-stage statement. No lower bound for D_{II} independent of the data is asserted. □

The exact equality improves the bookkeeping over a bare sign test: loss through omitted ports and exterior absorbers is preserved as a positive factor. It handles the missing corner directions of `imp:schur-interface`, even when the coupling to one of them is arbitrarily small. It does not imply that its inverse is polynomially bounded under the required weighted law.

Proposition 4.2 (Forcing and inexact reconstruction)**.** *For* $Dx = b$, *define* $b_e = b_R - D_{RI}D_{II}^{-1}b_I$. *For any proposed* $\widehat{x} = (\widehat{x}_R, \widehat{x}_I)$, *let* $r_I = b_I - D_{IR}\widehat{x}_R - D_{II}\widehat{x}_I$ *and* $r_e = b_e - D_e\widehat{x}_R$. *Its full residual is exactly*

$$b - D\widehat{x} = \begin{pmatrix} r_e + D_{RI}D_{II}^{-1}r_I \\ r_I \end{pmatrix}. \tag{4.3}$$

The exact solution, when defined, has $x_I = D_{II}^{-1}b_I - D_{II}^{-1}D_{IR}x_R$.

Proof. Solve the definition of r_I for $\widehat{x}_I$ and substitute in $b_R - D_{RR}\widehat{x}_R - D_{RI}\widehat{x}_I$. The terms without r_I give r_e, and the remaining term has the displayed positive sign. Setting both residual blocks to zero gives the solution formula. □

Thus a report of a tiny r_e alone is not an acceptable source-response certificate. At a weak missing-direction pivot the transported term can be much larger. Direct multiplication by the original sparse D is a useful independent check of the reconstructed residual.

Theorem 4.3 (Quadratic passivity budget for an inexact pivot)**.** *Let* $D = \left(\begin{smallmatrix} A & C \\ C^\top & B \end{smallmatrix}\right)$ *be complex symmetric with* $W = -\operatorname{Im} D \succeq 0$. *For any approximate pivot action* $\widehat{X}$, *set*

$$R = C^\top - B\widehat{X}, \qquad \widehat{D}_e = A - C\widehat{X}, \qquad \widehat{J} = \begin{pmatrix} I \\ -\widehat{X} \end{pmatrix}.$$

Without assuming B *invertible one has*

$$-\operatorname{Im} \widehat{D}_e = \widehat{J}^* W \widehat{J} - \operatorname{Im}(\widehat{X}^* R). \tag{4.4}$$

If the full absorption inequality $W \succeq \operatorname{diag}(W_0, \beta I)$ *holds with* $\beta > 0$, *then*

$$-\operatorname{Im} \widehat{D}_e \succeq W_0 - \frac{R^*R}{4\beta}. \tag{4.5}$$

In particular, if $W_0 \succeq \alpha I$ *and* $\delta = \alpha - \|R\|^2/(4\beta) > 0$, *the approximate Schur matrix is invertible with* $\left\|\widehat{D}_e^{-1}\right\| \le \delta^{-1}$. *Under the weaker inequality* $W \succeq \operatorname{diag}(W_0, 0)$ *one only obtains the general bound* $-\operatorname{Im} \widehat{D}_e \succeq W_0 - \left\|\widehat{X}\right\| \|R\| I$.

Proof. $D\widehat{J} = (\widehat{D}_e, R)^\top$, hence $\widehat{J}^* D \widehat{J} = \widehat{D}_e - \widehat{X}^* R$. Taking negative imaginary parts gives (4.4). Under the stronger absorption condition its right side is bounded below by $W_0 + \beta \widehat{X}^* \widehat{X} - \operatorname{Im}(\widehat{X}^* R)$. Complete the square:

$$\beta \widehat{X}^* \widehat{X} - \operatorname{Im}(\widehat{X}^* R) = \beta \left(\widehat{X} + \frac{iR}{2\beta} \right)^* \left(\widehat{X} + \frac{iR}{2\beta} \right) - \frac{R^* R}{4\beta}.$$

This proves (4.5). If the lower bound is $\delta I > 0$, then $\left\| \widehat{D}_e v \right\| \|v\| \geq |\operatorname{Im}(v^* \widehat{D}_e v)| \geq \delta \|v\|^2$. In finite square dimension this implies the inverse estimate. Finally $\left\| \operatorname{Im}(\widehat{X}^* R) \right\| \leq \left\| \widehat{X} \right\| \|R\|$ gives the weaker assertion. □

This residual refinement, proved directly above, certifies a structural property of an inexact Schur matrix with a quadratic rather than linear residual debit. It requires absorption on the eliminated coordinates; it cannot be applied with an invented positive β to the source's unabsorbed missing directions. A retained diagonal block bound alone is not the full block inequality. Also $\widehat{D}_e$ need not be exactly symmetric; reciprocity must be checked separately. If B is invertible, the forward error is $\widehat{D}_e - D_e = CB^{-1}R$, so passivity and forward accuracy remain different questions.

The necessity of the full absorption inequality has a sharp finite counterexample. Take

$$D = \begin{pmatrix} 1-i & -i \\ -i & -i \end{pmatrix}, \qquad W = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \succeq 0.$$

The retained absorption entry is one, and both D and its eliminated block are invertible. Nevertheless the exact Schur complement is 1, with zero absorption. Thus $W_{RR} \succeq W_0$ cannot replace $W \succeq \operatorname{diag}(W_0, 0)$.

5 Heterogeneous Passive Port Closure

The common phase of `mat:closure` is generalized here to distinct port phases and, algebraically, to a contractive matrix termination. No probabilistic independence is needed for this finite identity. The transpose convention is consequential: for $S = i \left(\begin{smallmatrix} 3/5 & 4/5 \\ 4/5 & -3/5 \end{smallmatrix} \right)$ and $z = i/2$, correct common-phase closure gives $i/7$. Replacing only $T^\top$ by T^* gives $37i/35$, whose modulus exceeds one strictly inside the disk. Both values follow by scalar substitution. The latter formula is not a passive representation.

Theorem 5.1 (Feedback and the exact loss decomposition)**.** *Let $S = \left(\begin{smallmatrix} A & T \\ C & B \end{smallmatrix} \right)$ be a contraction on retained and closed port spaces, with $\mathcal{D} = I - S^* S \succeq 0$. For a square closed-port matrix Z with $\|Z\| < 1$, define*

$$Q_Z = (I - BZ)^{-1} C, \quad F_Z = A + TZQ_Z, \quad V_Z = \begin{pmatrix} I \\ ZQ_Z \end{pmatrix}. \tag{5.1}$$

Then

$$I - F_Z^* F_Z = V_Z^* \mathcal{D} V_Z + Q_Z^* (I - Z^* Z) Q_Z \succeq 0. \tag{5.2}$$

The formula remains valid for $\|Z\| \leq 1$ whenever $I - BZ$ is invertible. If $\mathcal{D} = K^* K$, a loss factor for the terminated matrix is the vertical stack of KV_Z and $(I - Z^* Z)^{1/2} Q_Z$. For symmetric S and diagonal $Z = \operatorname{diag}(z_j)$, F_Z is symmetric and jointly holomorphic on the open polydisk $|z_j| < 1$.

Proof. $\|BZ\| < 1$ makes the inverse in (5.1) a convergent Neumann series. The feedback equation is $C + BZQ_Z = Q_Z$, so

$$SV_Z = \begin{pmatrix} F_Z \\ Q_Z \end{pmatrix}.$$

Evaluate $V_Z^*(I - S^*S)V_Z$ using these two block columns and rearrange. This proves (5.2) and the factorization. At an invertible boundary point exactly the same multiplication works. In the symmetric case $C = T^\top$ and $Z(I - BZ)^{-1} = (I - ZB)^{-1}Z$; its transpose is itself because $B^\top = B$ and $Z^\top = Z$. This proves symmetry. Locally uniform convergence of the Neumann series on compact subpolydisks proves joint holomorphy. □

5.1 Physical meaning and composition

For the physical lossless S of (1.2), a diagonal termination with $z_j = (w_j + i)/(w_j - i)$ corresponds to

$$D_{\boldsymbol{w}} = D + Y_C \operatorname{diag}(w_j + i) Y_C^\top. \tag{5.3}$$

For real finite w_j, this closes the selected absorbers and restores the real shifts $w_j\sigma_j$ at their sites. For $\operatorname{Im} w_j < 0$ it leaves a passive termination. Indeed the inverse Cayley map is $w_j = -i(1+z_j)/(1-z_j)$, and maps the disk to the lower half-plane. Woodbury with diagonal $\operatorname{diag}(w_j + i)$, initially where all its factors are defined, gives F_Z in (5.1); the equality then holds as a rational identity wherever the physical inverse exists. Repeated physical sites cause no change, because the columns of Y_C were never merged. The value $z_j = 1$ denotes an infinite shift in this parameterization, not an allowed finite w_j.

With unitary S, (5.2) says exactly how much norm is absorbed by nonunit-modulus terminations. With a contractive Schur or reduced response, it also records the preexisting loss $\mathcal{D}$. Successive closures give a new contractive matrix and its explicit stacked loss factor, so the theorem can be reapplied. Simultaneous and successive formulas agree whenever their elimination pivots exist: both solve the same linear feedback equations, whose full solution is unique. This is a compositional certificate, not a claim that every possible intermediate pivot is nonsingular.

Lemma 5.2 (Dark modes in common-phase closure)**.** *Suppose S is unitary and $Z = zI$. Let U be the span of all eigenvectors of B with eigenvalue of modulus one. Then U reduces B, $T|_U = 0$ and $U^*C = 0$ (where U also denotes an orthonormal inclusion). On $U^\perp$ all eigenvalues of B have modulus less than one. The common-phase transfer F_z therefore has an analytic continuation to a neighborhood of the closed unit disk, despite possible singularities of the full middle matrix on its boundary.*

Proof. For $Bu = \lambda u$, $|\lambda| = 1$, contractivity implies $(I - B^*B)u = 0$ and hence $B^*u = \bar{\lambda}u$. Thus its eigenspace reduces B. The unitarity relations $T^*T + B^*B = I$ and $CC^* + BB^* = I$ give $Tu = 0$ and $C^*u = 0$. The span of such reducing eigenspaces reduces B as well; there are no remaining unit-modulus eigenvalues on the orthogonal complement. In finite dimension, $(I - zB|_{U^\perp})^{-1}$ is analytic on a neighborhood of $|z| \le 1$. The U part is absent from the sandwich, proving the claim. □

This is a transfer-function cancellation, not invertibility of the physical site matrix at every real closing value. Nor does it supply a uniform distance of the observable eigenvalues from the circle over exterior data or volume. The single common-phase dark-space argument is not automatically a simultaneous multivariable boundary regularity assertion.

Theorem 5.3 (Inexact feedback certificate)**.** *For the data of Theorem 5.1, take any $\widehat{Q}$ and form the* original-equation *residual and response*

$$R = C - (I - BZ)\widehat{Q}, \qquad \widehat{F} = A + TZ\widehat{Q}, \qquad \widehat{V} = \begin{pmatrix} I \\ Z\widehat{Q} \end{pmatrix}.$$

For either operator or Hilbert–Schmidt norm, if $\|BZ\| \le r < 1$,

$$\left\| F_Z - \widehat{F} \right\| \le \frac{\|TZ\|}{1-r} \|R\|. \tag{5.4}$$

More generally replace $(1-r)^{-1}$ by any certified bound for $\|(I-BZ)^{-1}\|$. The exact loss residual identity is

$$\begin{aligned} I - \widehat{F}^*\widehat{F} = \widehat{V}^*\mathcal{D}\widehat{V} + \widehat{Q}^*(I - Z^*Z)\widehat{Q} \\ + \widehat{Q}^*R + R^*\widehat{Q} + R^*R. \end{aligned} \tag{5.5}$$

In particular an arbitrary approximate solve is not automatically passive. Its response error δ gives $\left\|\widehat{F}\right\| \le 1 + \delta$.

Proof. Subtract the two feedback equations to obtain $Q_Z - \widehat{Q} = (I - BZ)^{-1}R$. Multiplication by TZ and the Neumann bound prove (5.4). Next $S\widehat{V} = (\widehat{F}, \widehat{Q} + R)^\top$. Expand $\widehat{V}^*\mathcal{D}\widehat{V}$ and rearrange to get (5.5). Finally use $\|F_Z\| \le 1$ and the triangle inequality. □

There is no claim of stability at all boundary phases from a disk bound. Near a small middle singular value the error bound degrades; dark-mode removal or a certified absolute margin must be supplied.

6 Residual Certificates in the Observed Physical Norm

Theorem 6.1 (Absorbed-source residual bound)**.** *Let D be invertible, $W = -\operatorname{Im} D \succeq YY^*$, and let $DX = B_0$ be a finite multi-source system. For a candidate $\widehat{X}$, set $R = B_0 - D\widehat{X}$. If*

$$R = W^{1/2}Z_0, \tag{6.1}$$

then

$$\left\| Y^*(X - \widehat{X}) \right\| \le \|Z_0\|. \tag{6.2}$$

The assertion holds in operator and Hilbert–Schmidt norms. The minimum-norm choice is $Z_0 = W^{\dagger/2}R$ when every column of R lies in $\operatorname{ran} W^{1/2}$. For a splitting $R = W^{1/2}Z_0 + R_\perp$ one instead has

$$\left\| Y^*(X - \widehat{X}) \right\| \le \|Z_0\| + \left\| Y^*D^{-1}R_\perp \right\|. \tag{6.3}$$

The last term cannot be discarded from a sparse-port problem.

Proof. For one right-hand side $r = W^{1/2}z$, let $e = D^{-1}r$ and $t = \left\|W^{1/2}e\right\|$. Then

$$t^2 = -\operatorname{Im}(e^*De) = -\operatorname{Im}(e^*W^{1/2}z) \le t\|z\|.$$

Thus $t \le \|z\|$, including $t = 0$. Since $YY^* \preceq W$, $\|Y^*e\| \le t$. Apply this to every linear combination of columns for the operator norm, and to the columns separately and sum squares for the Hilbert–Schmidt norm. The pseudoinverse chooses the representative orthogonal to $\ker W^{1/2}$; all others add a vector in that kernel and cannot improve either norm. Linearity and the triangle inequality give (6.3). □

This estimate is independent of $\|D^{-1}\|$, but only in the declared residual metric. It is useful for the source's commutator residuals: changing cutoff edges have both endpoints at probe sites, so those residuals are represented by absorbed sources. It is also useful for projected Schur residuals with a positive retained absorption floor. It is not an unconditional error bound for every sparse-port Galerkin residual.

Corollary 6.2 (Uniform absorption and scattering error)**.** *If $W \succeq \eta I$ with $\eta > 0$, every residual satisfies the range condition and $\left\|Y^*(X - \widehat{X})\right\| \le \eta^{-1/2} \|R\|$. For $B_0 = Y$ and $\widehat{S} = I + 2iY^*\widehat{X}$ this gives*

$$\left\|S - \widehat{S}\right\| \le 2\left\|W^{-1/2}R\right\| \le 2\eta^{-1/2}\|R\|.$$

If instead a source-specific projected bound $\|\Pi^* D^{-1}\Pi\| \le K$ is supplied, $Y = \Pi J$ and $R = \Pi Z_0$, then $\left\|S - \widehat{S}\right\| \le 2\|J\| K \|Z_0\|$.

Proof. The first assertion follows from $\left\|W^{-1/2}\right\| \le \eta^{-1/2}$ and Theorem 6.1. The second follows by subtracting the definitions of the scattering matrices. For the last assertion, $Y^*D^{-1}R = J^*\Pi^*D^{-1}\Pi Z_0$. □

The last alternative uses exactly the type of estimate in `screen:projected`; it does not prove that estimate or supply its possibly large constant K. Adding uniform absorption solely to make a certificate easier changes the problem and must be separately removed; no such silent regularization is permitted here.

Proposition 6.3 (Certified energy comparison)**.** *For any two matrices $S, \widehat{S}$ on the same port set and real vector data of diameter $D_f = \max_{a,j}|f_j - f_a|$, the pairwise energies satisfy*

$$\left|\sqrt{\mathcal{Q}_S(f)} - \sqrt{\mathcal{Q}_{\widehat{S}}(f)}\right| \le \frac{D_f}{\sqrt{2}}\left\|S - \widehat{S}\right\|_{\mathrm{HS}}. \tag{6.4}$$

Thus an error bound δ in Hilbert–Schmidt norm encloses the true energy between $\max(0, \sqrt{\mathcal{Q}_{\widehat{S}}} - D_f\delta/\sqrt{2})^2$ and $(\sqrt{\mathcal{Q}_{\widehat{S}}} + D_f\delta/\sqrt{2})^2$.

Proof. View $((f_j - f_a)S_{ja})_{ja}/\sqrt{2}$ as one vector in the Euclidean direct sum over ordered port pairs and data components. Its norm is $\sqrt{\mathcal{Q}_S}$. The reverse triangle inequality and $|f_j - f_a| \le D_f$ prove the claim. □

This is a finite-data bound. Affine data on growing tori have growing diameter, and operator-norm error alone incurs rank factors in conversion to Hilbert–Schmidt norm. No dimension-independent boundary moment estimate is inferred from this proposition.

7 Passive Reduction and Coordinate Metrics

Theorem 7.1 (Real Galerkin port representation)**.** *Assume $D = D^\top$, Y real, and $W = -\operatorname{Im} D \succeq YY^\top$. Let $V \in \mathbb{R}^{n\times k}$ have full column rank and suppose $D_r = V^\top DV$ is invertible. Put*

$$Y_r = V^\top Y, \quad W_r = V^\top WV, \quad S_r = I + 2iY_r^\top D_r^{-1}Y_r.$$

Then S_r is symmetric contractive. Its exact loss is

$$I - S_r^*S_r = 4Y_r^*D_r^{-*}(W_r - Y_rY_r^*)D_r^{-1}Y_r \succeq 0. \tag{7.1}$$

It is unitary if $W = YY^\top$. The lifted response $X_r = VD_r^{-1}Y_r$ has full residual $R_r = Y - DX_r$ and $V^\top R_r = 0$. Whenever its residual satisfies a certificate in Section 6, that certificate bounds $S - S_r$. All statements are invariant under $V \mapsto VM$ for real invertible M, with the coefficient metric transformed accordingly.

Proof. $W_r - Y_r Y_r^* = V^\top (W - YY^\top)V \succeq 0$, and D_r is complex symmetric. With $K_r = Y_r^* D_r^{-1} Y_r$, the inverse-difference identity gives $\operatorname{Im} K_r = Y_r^* D_r^{-*} W_r D_r^{-1} Y_r$. Expanding $(I + 2iK_r)^*(I + 2iK_r)$ gives (7.1). Symmetry follows by transpose. If $W = YY^\top$, the right side vanishes. Direct multiplication gives $V^\top R_r = Y_r - D_r D_r^{-1} Y_r = 0$. Finally $(M^\top D_r M)^{-1} = M^{-1} D_r^{-1} M^{-\top}$, so the lifted response and port scattering are unchanged by the basis change. □

The Gram matrix is $M_V = V^\top V > 0$. The coefficient operator $M_V^{-1} D_r$ represents the projected physical action, with physical norm $\|c\|_{M_V}^2 = c^* M_V c$. Its dissipativity is measured by $c^* D_r c$, not by the Euclidean imaginary part of $M_V^{-1} D_r$. Orthonormalization with $V M_V^{-1/2}$ gives the same response. Real coordinates preserve transpose symmetry; complex trial bases with adjoint compression still preserve passivity, but generally lose reciprocity in this simple form.

Invertibility of D does not ensure that of D_r. For example, $D = \left(\begin{smallmatrix} 0 & 1 \\ 1 & -i \end{smallmatrix}\right)$ is invertible passive, but $V = e_1$ gives $D_r = 0$. Moreover $V^\top R_r = 0$ is not $R_r = 0$ and does not imply an absorbed-source representation for R_r. These are mandatory rejection conditions, not small numerical defects.

The realized gain is a precise separation of *structure* from *accuracy*: a reduced response can be provably passive while inaccurate, and an accurate computed response can slightly violate passivity because of its solve residual. Equations (7.1), (6.2), and (6.4) test different obligations and should all be retained. No claim is made that a particular trial space reduces cost or preserves the source's weighted probability law.

8 An Exact Cauchy Energy Decrement

The source's conditional comparison discards a negative square. Keeping it and evaluating the ideal Cauchy second moment gives the following strictly more informative finite identity.

Theorem 8.1 (Barycentric closure identity)**.** *Let S be a finite symmetric unitary matrix, a a port, and f_j fixed real vector data, all measurable before the closing phase is sampled. Set $b = S_{aa}$, $u = S_{\widehat{a},a}$, and $d = 1 - |b|^2 = \sum_{j \neq a} |u_j|^2$. If $d > 0$, define*

$$\bar{f}_a = d^{-1} \sum_{j \neq a} |u_j|^2 f_j.$$

Close a with an independent Haar phase z on the unit circle, so $S_z = A + \gamma u u^\top$, $\gamma = z/(1 - bz)$. Then

$$\mathbb{E}_z \mathcal{Q}_{S_z}(f_{\widehat{a}}) = \mathcal{Q}_S(f) - d\,|\bar{f}_a - f_a|^2. \tag{8.1}$$

When $d = 0$ the matrix on the other ports is unchanged and the decrement is defined to be zero. Haar phase is equivalent to standard Cauchy w in $z = (w + i)/(w - i)$.

Proof. For scalar data subtract f_a, so that the removed datum is zero, and put $F = \operatorname{diag}(f_j - f_a : j \neq a)$. The trace energy formula and unitarity of S_z give the exact identity

$$\mathcal{Q}_{S_z} - \mathcal{Q}_S = -2 \operatorname{Re}\{\gamma u^\top F A^* F u\} - |\gamma|^2 (u^* F u)^2. \tag{8.2}$$

Indeed expand $\operatorname{tr}(F^2 - F S_z F S_z^*)$; the pre-closure trace is $\operatorname{tr}(F^2 - FAFA^*)$ because the deleted datum is zero. The two linear terms are conjugates. The quadratic trace is $(u^* F u)^2$, a nonnegative

real square. For $|b| < 1$, the uniformly convergent series $\gamma = \sum_{k\ge 0} b^k z^{k+1}$ has mean zero and, by Fourier orthogonality,

$$\mathbb{E}_z|\gamma|^2 = \sum_{k\ge 0} |b|^{2k} = d^{-1}.$$

Also $u^*Fu = d(\bar{f}_a - f_a)$. Substitute into (8.2). For vector data sum over real components, yielding the Euclidean squared norm. If $d = 0$, unitarity gives $u = 0$, so no retained scattering entry changes. Finally $w = \cot(t/2)$ for $z = e^{it}$ transforms normalized Haar measure to $\mathrm{d}w/[\pi(1 + w^2)]$. □

This identity distinguishes zero expected cost from strict energy decrease. Equality holds exactly when $f_a = \bar{f}_a$ on a coupled port. For the symmetric two-port swap matrix $S = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and data $(0, 1)$, the initial energy is one and the retained one-port energy is zero: the decrement is exactly one, not just nonnegative. Thus the refinement is not merely a numerical-factor improvement in an upper bound.

Corollary 8.2 (Finite telescoping certificate)**.** *Close finitely many ports in an order chosen from the history of previous closures, using a fresh conditionally Haar phase at each step. Keep the surviving data equal to their original real values. Writing d_k and $\bar{f}_k$ for the current open system at step k, one has*

$$\mathbb{E}\mathcal{Q}_{\text{final}} + \sum_k \mathbb{E}\big[d_k|\bar{f}_k - f_{a_k}|^2\big] = \mathcal{Q}_{\text{initial}}.$$

For random initial data the right side is its expectation, assuming integrability. All terms are nonnegative.

Proof. Condition on the past, apply Theorem 8.1, and use the tower property. Sum the finite number of identities. For a fixed finite port set bounded data and unitarity ensure integrability; the stated integrability condition covers random initial data. □

8.1 The bounded-law boundary

For a general conditional closing law with finite $\mu = \mathbb{E}\gamma$ and $\nu = \mathbb{E}|\gamma|^2$, the same calculation gives

$$\mathbb{E}(\mathcal{Q}_{S_z} - \mathcal{Q}_S) = -2\,\mathrm{Re}(\mu u^\top FA^*Fu) - \nu(u^*Fu)^2 \tag{8.3}$$

componentwise. Since $|u^\top FA^*Fu| \le \|Fu\|^2$, discarding the last term recovers `mat:conditional-comparison`. For $d > 0$, γ is bounded on the circle, so the moments exist for every probability law there. The $d = 0$ case is handled directly and never by division.

Theorem 8.3 (Centered comparison for the actual conditional law)**.** *For the same open matrix, port, and pre-hidden real data as in Theorem 8.1, replace its Haar assumption by any conditional probability measure on the unit circle and let $d > 0$. Write $m = \bar{f}_a - f_a$ and, for each real data component ℓ, put*

$$F_\ell = \mathrm{diag}(f_{j,\ell} - f_{a,\ell}), \qquad v_\ell = (F_\ell - m_\ell I)u.$$

Then $u^*v_\ell = 0$ and the exact conditional identity is

$$\mathbb{E}_z(\mathcal{Q}_{S_z} - \mathcal{Q}_S) = -d|m|^2 - 2\,\mathrm{Re}\left(\mu \sum_\ell v_\ell^\top A^* v_\ell\right). \tag{8.4}$$

In particular, defining the normalized variance $V_a = d^{-1}\sum_{j\ne a}|u_j|^2|f_j - \bar{f}_a|^2$, one has

$$\mathbb{E}_z\mathcal{Q}_{S_z} \le \mathcal{Q}_S - d|\bar{f}_a - f_a|^2 + 2|\mu|dV_a. \tag{8.5}$$

Thus $|\bar{f}_a - f_a|^2 > 2|\mu|V_a$ certifies strict expected decrease for this open configuration. No independence of the closing phase from previously exposed data beyond its specified conditional law is required. At $d = 0$ all terms are defined to be zero.

Proof. For every $|z| = 1$ and $\gamma = z/(1 - bz)$, direct multiplication gives

$$d|\gamma|^2 = 1 + 2\,\mathrm{Re}(b\gamma), \qquad d\nu = 1 + 2\,\mathrm{Re}(b\mu).$$

For one component, $u^*Fu = dm$ by the definition of m, so $u^*v = 0$. The off-diagonal identity from $S^*S = I$ is $A^*u = -b\bar{u}$. Since A is symmetric, A^* is symmetric too, and transposition gives $u^\top A^* = -bu^*$. Consequently, expanding $Fu = mu + v$ and using $m \in \mathbb{R}$,

$$u^\top F A^* F u = v^\top A^* v - bdm^2;$$

both mixed terms vanish by $u^*v = 0$, and $u^\top A^* u = -bd$. Substitute this expression and $d\nu = 1 + 2\,\mathrm{Re}(b\mu)$ into (8.3). The terms involving $\mathrm{Re}(b\mu)dm^2$ cancel exactly and leave $-dm^2 - 2\,\mathrm{Re}(\mu v^\top A^* v)$. Sum over components to get (8.4). Contractivity of A gives $|v^\top A^* v| \le \|v\|^2$, and $\sum_\ell \|v_\ell\|^2 = dV_a$, proving (8.5). When $d = 0$, $u = 0$ and the retained system does not change, as already noted. □

The centered estimate is at least as sharp as the source's upper comparison for every fixed open configuration. Indeed the weighted variance decomposition gives

$$\mathcal{M}_a(f) = \sum_{j \ne a} |u_j|^2 |f_j - f_a|^2 = dV_a + d|m|^2.$$

Relative to $\mathcal{Q}_S + 2|\mu|\mathcal{M}_a(f)$, the new upper bound is smaller by $(1 + 2|\mu|)d|m|^2$. In particular, if the retained data are constant at a value different from f_a, then $V_a = 0$ and the exact decrement $d|m|^2$ holds for *every* conditional closing law, not just ideal Cauchy.

For the source's cutoff law, its existing bound $|\mu| \le C\sqrt{\sigma}$ on $|g| > \sqrt{\sigma}$ can be inserted directly into (8.5); elsewhere use its uniform $|\mu| \le C$. Thus this is an actual bounded-law local refinement, not a proposal to replace the physical uniform distribution by an unbounded one. It preserves the requirement that all open data and the chosen f be measurable without the hidden draw. No averaged lower bound for $d|m|^2$ over the hierarchy is proved here. The source's cutoff, likelihood normalization, weighted transmission, and limiting capacity argument remain necessary. Finite strict decrease is not a proof of vanishing limiting affine energy.

Corollary 8.4 (Finite centered error ledger)**.** *For finitely many predictable closures with arbitrary conditional circle laws, let d_k, m_k, V_k, μ_k be the current quantities above and $\tau_k = \sum_\ell v_{k,\ell}^\top A_k^* v_{k,\ell}$. Keep the surviving data equal to their original real values and assume integrability of all displayed terms. Then*

$$\begin{aligned}\mathbb{E}\mathcal{Q}_{\mathrm{final}} + \sum_k \mathbb{E}[d_k|m_k|^2] &= \mathbb{E}\mathcal{Q}_{\mathrm{initial}} - 2\sum_k \mathbb{E}\,\mathrm{Re}(\mu_k\tau_k) \\ &\le \mathbb{E}\mathcal{Q}_{\mathrm{initial}} + 2\sum_k \mathbb{E}[|\mu_k| d_k V_k].\end{aligned}$$

At a decoupled port its ledger entries are zero. This is a finite identity and bound, with no lower bound on the accumulated mismatch.

Proof. Condition on the history at each closure, use (8.4), and apply the tower property. The energies telescope; $|\tau_k| \le d_k V_k$ gives the inequality. □

9 The Conditional Infinite-Volume Bridge

Theorem 9.1 (Summable columns, with the input exposed)**.** *Fix $h > 0$, a real E and a site x. Couple centered odd finite tori to one infinite potential sample. Suppose their real inverses exist almost surely and, for some $0 < q < 2$, $a > 0$, and $C < \infty$, all sufficiently large N satisfy*

$$\mathbb{E} \sum_{y \in \Lambda_N(x)} e^{a|y-x|_\infty} |(H_N - E)^{-1}_{yx}|^q \le C. \tag{9.1}$$

Then almost surely there exists $u \in \ell^2(\mathbb{Z}^2)$ such that $(H_v - E)u = \delta_x$, and the site spectral measure obeys

$$\Gamma_x(E) = \int_{e \neq E} (e - E)^{-2} \, \mathrm{d}\mu_x(e) < \infty, \qquad \mu_x(\{E\}) = 0. \tag{9.2}$$

Proof. Let Z_N denote the sum in (9.1). Fatou gives $\mathbb{E} \liminf Z_N \le C$, so its liminf is finite almost surely. For such a sample choose a subsequence with bounded Z_N. Every fixed coordinate of its inverse column is bounded by that common bound to the power $1/q$. Diagonal extraction over the countable lattice gives coordinatewise convergence along a further subsequence. Fatou for counting measure yields $\sum_y e^{a|y-x|} |u_y|^q < \infty$. This sequence is bounded, and for $q < 2$, $\sum |u_y|^2 \le \|u\|_\infty^{2-q} \sum |u_y|^q < \infty$. Each fixed nearest-neighbor equation eventually avoids the periodic boundary, so it passes to $(H_v - E)u = \delta_x$. The spectral theorem gives $\mathrm{d}\mu_x(e) = (e - E)^2 \, \mathrm{d}\mu_u(e)$. It follows that $\mu_x(\{E\}) = 0$ and $\Gamma_x(E) \le \mu_u(\mathbb{R}) = \|u\|^2$. □

The input follows conditionally from (2.1), choosing an exponential weight smaller than its decay rate and summing the $O(r)$ sites in each planar max-norm shell. All constants are then fixed before N grows. The proof is also valid in other finite dimensions with the corresponding polynomial shell count, or for any summable column majorant. That generality does not extend the source's two-dimensional proof of the hypothesis.

9.1 Why this is not yet pure point

For each fixed E, the conclusion holds on an E-dependent probability-one event. Eigenvalues are random; an exceptional Lebesgue-null set of energies can still carry all spectral mass. One therefore cannot conclude purity merely by saying "the resolvent column is square summable almost everywhere".

The correct additional route, used by the source, is as follows. The true spectral kernel $\mu_x(v)$ is jointly measurable: bounded operator functional calculus first treats continuous functions by polynomial approximation, and monotone approximation treats the nonnegative inverse-square kernel. Fubini then gives a.e. exterior and a.e. base value at x with finite Γ_x for Lebesgue-a.e. energy. At this fixed exterior choose one such base value. For the rank-one family $H_a = H_0 + a|\delta_x\rangle\langle\delta_x|$,

$$F_a(z) = \frac{F_0(z)}{1 + aF_0(z)}, \qquad \int_{\mathbb{R}} \mu_a \, \mathrm{d}a \le \mathrm{Leb}\,.$$

The averaging inequality follows by integrating $\operatorname{Im} F_a(E + i\eta)$ in a, obtaining π, then testing against nonnegative continuous functions and applying Fatou. At a base energy with finite Γ_x, the spectral inverse vector exists; if $1 + aF_0(E + i0) = 0$ it is a nonzero eigenvector of H_a. At other such energies the boundary imaginary part vanishes and the corresponding scalar spectral mass is zero, by the usual measure differentiation argument. Averaging removes the remaining Lebesgue-null exceptional energies. This is the classical Simon–Wolff mechanism [3], reproduced in detail in `spec:averaging`; it is not a Ram discovery.

Finally one must average the allowed v_x interval, remove its two fixed spectral endpoints by the same averaging inequality, and intersect over countably many x. The spectrum inclusion $[-4-h, 4+h]$ follows from the norm bound; equality uses independent nearly constant boxes and cutoff plane waves. Neither that Weyl sequence construction nor purity of a single cyclic subspace can replace purity for all site spectral measures.

10 Verification and Evidence

10.1 What constitutes a certificate

Theorems in Sections 4–8 are finite algebraic or finite conditional-expectation assertions with proofs given here. Their inputs are exact matrices, positivity, explicit invertibility domains, declared metrics, and the relevant conditional law. A computational implementation must additionally bound rounding in the residual and certify positivity or singular-value margins. Ordinary floating-point tests are checks, not validated interval certificates or proofs for all dimensions.

The local check program is `_audit/M524-author-checks.py`; its bounded results are written to `_audit/M524-author-check-results.json`. It tests rational Ward, Schur, forcing, feedback-loss and inexact-feedback identities; seeded finite lossy and lossless examples; nonorthogonal real trial bases; repeated site ports; the absorbed residual bound; the dark-mode cancellation example; and the exact two-port decrement. A finite circle quadrature is a diagnostic for the Cauchy identity, not its proof. No rare-cell Monte Carlo, localization-length computation, speed benchmark, or all-energy numerical conclusion is claimed.

10.2 Executed bounded checks

All twelve exact Gaussian-rational checks and all 24 seeded finite numerical trials passed. The largest site and port dimensions were seven and nine, respectively; these are finite matrix tests, not a simulation of the rare-cell hierarchy. The tests included both lossless and lossy systems, repeated site ports, and nonorthogonal real trial bases. A 2048-point circle quadrature checked the ideal Cauchy decrement on one three-port example. A five-atom non-Haar circle law and a 4097-point compactly supported Cauchy quadrature checked the centered conditional identity and bound. The latter is a quadrature diagnostic, not a validated integral enclosure. The largest numerical identity discrepancy was 2.85×10^{-14}, below the specified 5×10^{-11} tolerance. NumPy version 2.3.5 and Python version 3.12.14 were used; exact tests used rational real and imaginary parts.

A separate targeted mathematical review found no consequential error under the stated assumptions in the Schur loss and quadratic residual budget, heterogeneous closure and residual signs, absorbed-source norm bound including its range condition, real Galerkin loss, Cauchy decrement and finite telescoping, and the conditional column bridge. The centered-law refinement was independently derived as shown above and replayed against the original energy definition. A separate exact rational-complex replay passed seven fixtures, covering every port in three- and four-port symmetric unitary matrices, two real data components, and a biased three-point phase law. It checked the phase moment relation, closure unitarity, centered algebra, exact expected energy, and the two-sided error bound. These finite replays supplement the proof and do not replace it. The separate source audit likewise found no defect within its declared bounded scope. Neither review certifies the complete bootstrap or the headline localization assertion. Local evidence records distinguish proof, exact finite verification, floating-point diagnostics, and unverified global dependencies.

11 Comparisons, Limitations, and Research Directions

11.1 Public mathematical context

The public literature was checked by public mathematical topics and titles only; private corpus text was not included in queries. The finite-volume fractional-moment criteria of Aizenman, Schenker, Friedrich and Hundertmark [2] place an explicit smallness criterion before localization. Our finite feedback and residual certificates are not such a probabilistic criterion. Simon–Wolff [3] supplies the rank-one spectral conversion, not the all-disorder planar input.

Hurtado's 2026 paper [4], Theorems 1.4–1.6, treats independent uniformly bounded potentials with uniformly positive variance and obtains localization in a bottom interval. Its normalization is $-\Delta + V$, rather than the adjacency convention (1.1); interval comparisons require the corresponding shift/sign convention. It does not independently verify the pinned all-energy assertion. Black, Drogin and Hernández [5] study Gaussian weak disorder and diffusive resolvent/time scales; that scaling regime and disorder law differ from fixed-h pure point spectrum for uniform variables. No conflict or equivalence is inferred merely from the word "diffusion".

Structure-preserving Galerkin reduction has established public precedents, including Egger and collaborators [6]. Our real-congruence proof is elementary passive linear algebra tailored to the source's complex symmetric port convention. The contribution is its explicit combination with original-equation residuals, heterogeneous termination loss, and energy comparison, not a claim to have originated passivity-preserving reduction.

11.2 Advantages and costs

The exact loss factor survives nested elimination and separates unobserved absorption from numerical error. The heterogeneous closure formula supports unequal physical shifts without merging duplicated site channels. The absorbed residual estimate controls the requested response without a global inverse margin when its range hypothesis is met. The Cauchy decrement identifies exactly which data mismatch produces a strict local energy decrease.

The costs are real: Schur fill-in can be dense; a loss factor may be large; rank deficiency makes pseudoinverse range checks indispensable; Galerkin compression can be singular even when the full matrix is not; and feedback conditioning can diverge near the circle. No runtime or memory improvement has been measured. Most importantly, finite certificates do not establish weighted impedance probabilities, a branch likelihood under an altered law, the vanishing of the limiting affine energy, or the uniform-in-volume fractional-moment estimate.

11.3 Concrete next steps

A meaningful next probabilistic step would average (8.5) under the actual cutoff law and compare the negative mismatch term with the centered weighted μ error. This requires hidden-value measurability and a lower bound on the barycentric mismatch over the relevant open configurations; the local conditional identity does not supply such a bound. A computational next step would implement interval residual evaluation and low-rank loss-factor compression with a certified positive remainder. It should benchmark the same physical operator, masks, tolerances, and boundary convention against direct sparse solves, counting preprocessing and failed certificates. A spectral next step would require summable, volume-uniform error and probability budgets compatible with (9.1); finitely many accurate tori do not supply them.

11.4 Realized contribution assessment

The internal C3 assessment credits the exact heterogeneous loss identity with compositional and inexact certificates, the observed-norm residual integration, the strict finite Cauchy decrement, and the centered-variance refinement valid for the actual conditional closing law. These are explicit proved refinements of the pinned finite-port interfaces and supply useful rejection tests. It does not credit the source's localization theorem, the known classical identities, unexecuted large-scale computations, or the proposed future probability gain as newly established work. A C4 or C5 assessment would require a stronger consequential advance at a source bottleneck than this report demonstrates. This is an internal evidence-based assessment, not external peer review.

A Exact Local Inputs and Read Scope

The study root is

`C:/Users/mural/OneDrive/Documents/OpenAI-722-math`

The source base below that root is

`_sources/openai-math/preprints/Pure-Point-Spectrum-for-the-Two-Dimensi`
`onal-Anderson-Model-at-Every-Positive-Disorder-September-23-2026`

The complete read set under that base is `README.md`, `build/paper.tex`, `build/references.tex`, and `build/sections/` followed by each of `introduction.tex`, `setup.tex`, `matrix.tex`, `resona nt.tex`, `impedance.tex`, `transmission.tex`, `hierarchy.tex`, `screening.tex`, `vanishing.tex`, `bootstrap.tex`, and `spectral.tex`. All their substantive text, proofs, mathematical figure definitions, and included bibliography were read. The source PDF is not present in this local directory; no claim of inspecting it is made.

The actual private passages are:

1. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat -5.tex`, lines 310–536: "Polynomial evaluation without eigendecomposition", "Coarsening, Lifts, and Schur Sources", "Topology Edits and Checked Low-Rank Reuse", and "Residuals, Error Ledgers, and Query Certificates", including all displayed proofs in that range.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01 d-physical-metrics.tex`, entire file: physical Gram definition, metric symmetry, residual decomposition, metric approximation, and finite-measurement limitations.

3. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr018/TR-rev-1 -old-tr018.tex`, lines 1155–1245: "Analytic Continuation of the Ram Transform in Order ν", its common-domain theorem and proof, counterexamples, and interpolation corollary. Other parts of that report are not claimed read.

4. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01f-s pectral-limits.tex`, entire file: nested graph-core pollution, strong versus norm resolvent convergence, stronger contracts, domain-aware inverse, and dual estimate.

File SHA-256 values, exact line counts and policy/helper provenance are recorded in `_audit/M52 4-author-inputs.json`. Hashes identify versions, not mathematical validity. The accompanying manifest binds the report, PDF, input ledger, check program and results, and review evidence to their recorded versions.

B Minimal Replay and Negative Tests

For rational finite tests, exact arithmetic can verify $D_e = X^* D X$, the Schur residual signs, Ward, and (5.5) without invoking probability. Positive-semidefinite inequalities in numerical examples remain floating-point diagnostics unless enclosed by validated arithmetic. The finite test sizes and seed are fixed in the check program.

Three negative tests are essential. First, with sparse absorption, an unabsorbed residual must not be assigned a zero error just because its pseudoinverse-weighted norm vanishes. Second, a unitary dark block can make $I - zB$ singular at a boundary phase while its observable transfer remains regular; test the transfer and the physical inverse as separate objects. Third, a reduced singular matrix must be rejected before a scattering formula is evaluated, even if the full matrix is invertible. These tests operationalize the private corpus's domain, metric and full-residual warnings.

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